SECOND LESSON

Quanta

The two pillars of twentieth-century physics – general relativity, of which I spoke in the first lesson and quantum mechanics, which I'm dealing with here – could not be more different from each other. Both theories teach us that the fine structure of nature is more subtle than it appears. But general relativity is a compact gem: conceived by a single mind, that of Albert Einstein, it's a simple and coherent vision of gravity, space and time. Quantum mechanics, or 'quantum theory', on the other hand, has gained unequalled experimental success and led to applications which have transformed our everyday lives (the computer on which I write, for example); yet more than a century after its birth it remains shrouded in mystery and incomprehensibility.

It's said that quantum mechanics was born precisely in the year 1900, virtually ushering in a century of intense thought. The German physicist Max Planck calculated the electric field in equilibrium in a hot box. To do this he used a trick: he imagined that the energy of the field is distributed in 'quanta', that is, in packets or lumps of energy. The procedure lead to a result which perfectly reproduced what was measured (and therefore must be in some way correct) but clashed with everything that was known at the time. Energy was considered to be something which varied continuously, and there was no reason to treat it as if it were made up of small building blocks. To treat energy as if it were made up of finite packages had been, for Planck, a peculiar trick of calculation, and he did not himself fully understand the reason for its effectiveness. It was to be Einstein once again who, five years later, came to understand that the 'packets of energy' were real.

Einstein showed that light is made of packets: particles of light. Today we call these 'photons'. He wrote, in the introduction to his article:

> It seems to me that the observations associated with blackbody radiation, fluorescence, the production of cathode rays by ultraviolet light, and other related phenomena connected with the emission or transformation of light are more readily understood if one assumes that the energy of light is discontinuously distributed in space. In accordance with the assumption to be considered here, the energy of a light ray spreading out from a point source is not continuously distributed over an increasing space but consists of a finite number of 'energy